

TEK4090

Introduction to Modern Control

Lecture 2: Classical Control

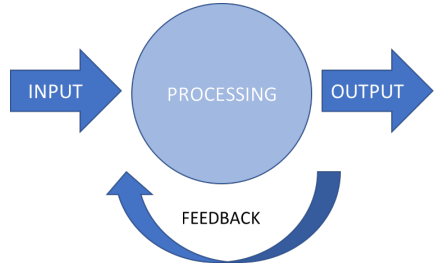
Mathias Hudoba de Badyn

August 20, 2025

What is Control?

Scientist

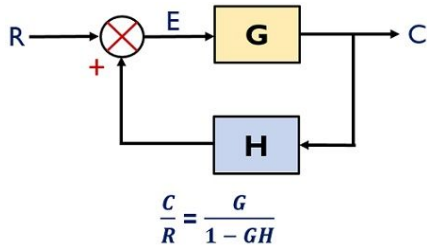
- ▶ The science of feedback in natural and engineered systems



What is Control?

Engineer

- The tools behind automation in engineered systems

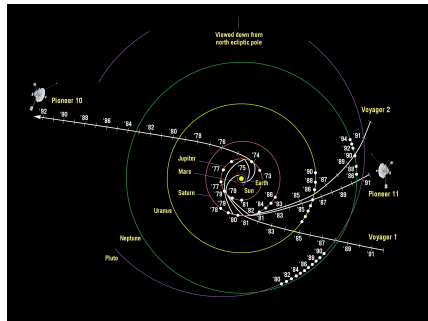


What is Control?

Mathematician

- Designing functions to ensure properties of solutions to differential equations

$$\dot{x} = f(x) + g(x) \cdot u(t) \quad (1)$$



What is Control?

Medieval Peasant

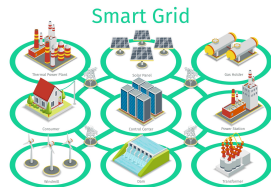
- Wizardry



Example: Smart Buildings



What are you all interested in?



Overview of the Course

Lecture	Topic	Application	Takeaways
1	Classical Control		Block Diagrams, transfer functions, stability
2	State Space Systems		Bode/Nyquist Plots, Controllability/Reachability
3	Optimal Control	Buildings, Power systems	Optimization, LQR, Ricatti Equation, MPC
4	Estimation	Spacecraft	Observability, Kalman filtering, duality, optimal estimator
5	Feedforward Control		System inversion, non-minimum phase systems
6	Robust Control	Hovering aircraft	PID, Root Locus/Loopshaping, Hinfinitiy Methods
7	System Identification	Health systems	PRBS, behavioural systems theory
8	Model-Predictive Control	Robotics	MPC, DeePC

- ▶ 8 lectures on theory, one on MATLAB
- ▶ Check the dates on syllabus - not every week has a lecture!
- ▶ overview of (almost) all control topics
- ▶ breadth
- ▶ not depth

The Team

Mathias Hudoba de Badyn



Mario Viridis



Canvas

1. Course syllabus is available on the Canvas page
2. There is a supplemental lecture on mathematics available on Canvas.
 - ▶ Please go through it and let me know if there are any topics there that you are unfamiliar with
 - ▶ Any topics that anyone is unfamiliar with will be covered in the lecture they are needed.
 - ▶ If no one tells me that they are unfamiliar with some of the topics, I will assume everyone is good
3. Please let me know ASAP if you cannot access the course Canvas page
4. (IT support is very slow at UiO)
5. Lectures will be recorded with Panopto and available here afterwards
6. In general, slides will be available on the Canvas page before the lecture

Matlab

Matlab is the gold standard in control engineering. We will be using Matlab for the course. You can install Matlab on Windows, Linux, and MacOS using the UiO-provided license.



- ▶ A tutorial will be provided on Matlab in the early stages of the course

Homeworks

50% of the final grade.

- ▶ There will be 4 homework assignments
- ▶ Each homework will correspond to 2 lectures
- ▶ If you submit all 4 assignments with a passing grade, the best 3 assignments will comprise 50% of your final grade
- ▶ You can work on assignments together, but your writeup must be your own
- ▶ Don't use ChatGPT to solve homework problems. A, it's not allowed per UiO policy, and B it's quite unreliable for whole problems.
 - ▶ It is, however, really good at explaining individual chunks of a problem
 - ▶ So, use it for 'tutorials' about individual topics, but don't use it for entire problems.

Final Project

The other 50% of the final grade.

- ▶ Write a 6-8 page research paper in L^AT_EX on a control topic of your choice
- ▶ Eg. pick an engineering system you like, do a deep dive into how to control it using tools from the course
- ▶ Eg. self-study a control theory topic beyond the scope of the course and formulate your own research question
- ▶ Application examples will be given throughout the course
- ▶ Slides from the applications will be made available within the first few weeks of the course
- ▶ **Details are on the canvas page**

Summary of information about the final project

Distributed Feedback Optimisation for Robotic Coordination

Antonio Terpin, Sylvain Fricker, Michel Perez, Mathias Hudoba de Badyn, Florian Dörfler

Abstract—Feedback optimisation is an emerging technique aimed at steering a system to an optimal steady state for a given objective function. We show that it is possible to employ this control strategy in a distributed manner. Moreover, we prove asymptotic convergence to the set of optimal configurations. To this scope, we show that exponential stability is needed only for the portion of the state that affects the objective function. This is showcased by driving a swarm of agents towards a target location while maintaining a target formation. Finally, we provide a sufficient condition on the topological structure of the specified formation to guarantee convergence of the swarm in formation around the target location.

I. INTRODUCTION

Feedback optimisation is an emerging technique aimed at steering a system to a trajectory computed online that is optimal with respect to a selected objective function, relying on little information on the controlled plant. In this paper, we apply feedback optimisation on distributed systems and extend convergence results to systems where a part of the state is only asymptotically but not exponentially stable. The results are then used to drive a swarm of robots in a given target formation towards a target location in a distributed manner.

A thorough review on the feedback-based optimisation methodology can be found in [1]. Remarkably, the plant dynamics need not be known. Instead, this approach relies only on the knowledge of the steady-state input-output sensitivity, allowing model-free optimisation and constraint handling [2], [3]. Furthermore, being a feedback-based approach results in the well-known advantages of feedback control, namely robustness to model mismatch and disturbances [4], [5]. A classical example is congestion control for cyber-networks, where source controllers and link dynamics are the modelled feedback loop [6]. In recent years, gradient projection algorithms and feedback optimisation gained traction within the field of power systems. The fact that

in time-varying optimisation are addressed in [13], [14], [15] reviews a broad class of algorithms for time-varying optimizations and shows how this can be applied to drive a single robot towards a target while avoiding collision. Convergence and stability analysis for regulation of linear time-invariant systems towards the optimal solution of a time-varying convex optimisation problem is studied in [16]. Constraints are included in [17], [18] and [19] extends to non-linear systems and non-convex problems.

Much of the work on robotic coordination for flocking relies on classical approaches inspired by the so-called Reynolds principles [20] and typically employs the concept of potential forces [21]. Along the lines of closed-loop optimisation for robotics coordination, feedback optimisation is used in [22] and [23] to learn generalised Nash Equilibrium in a non-cooperative game-theoretical setting. However, both of these require centralised or semi-decentralised algorithms. To the best of our knowledge, this paper is the first work that uses feedback optimisation for robotic coordination in a distributed manner. Our theoretical analysis on the asymptotic convergence of the closed-loop system builds instead on [24], where the authors quantify the required timescale separation to ensure stability and convergence of the interconnection of an exponentially stable plant and different schemes of feedback optimisation.

Our contributions are threefold. First, we show that it is possible to implement a feedback optimisation scheme in a distributed manner using robotic coordination as running example. Second, we prove the convergence of the swarm to the configuration that minimises the selected cost function. In particular, we build on [24] to show that exponential stability is required only for the portion of the state that affects the considered cost function whereas for the remainder of the state variables, only asymptotic stability is necessary. Furthermore, we derive conditions on the objective function coefficients and topology of the formation graph. There

- ▶ 6-8 page IEEE-format paper in $\text{\LaTeX}$
- ▶ Groups of 1–3 (expectations prorated)
- ▶ Sample projects are on Canvas
- ▶ Guide to writing control on Canvas
- ▶ Mix of coding and theory

Reading Assignments

Mathematics:

- ▶ **Linear Algebra:** Linear Algebra Done Right/Wrong (Ch 1–3, 5; Ch 1–2, 5)
- ▶ **Matrix Calculus:** The Matrix Cookbook
- ▶ **Complex Analysis:** Introduction to Complex Analysis (Ch 1–3)
- ▶ **Fourier & Laplace Transforms:** Feedback Systems (Ch 1–3, 5)

Classical Control:

1. Classical Control by Francis: Ch 3–5
2. Feedback Systems by Åstrom & Murray: Ch 8–11

Motivating Example

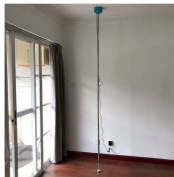
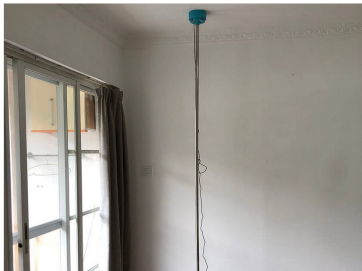
Motivating Example

Motivating Example

Ceiling Vibrator (Thumper): Loud Noisy Upstairs Neighbors Ultimate Solution

You know what?

When I started writing this article, I had just come through a dark time that caused me to suffer from mild depression and neurological debilitation until I got a **ceiling vibrator**. Also, others may call it ceiling thumper, ceiling vibration machine, or ceiling shaker.



Ceiling Vibrator System V2 Version:
Remote Control, Vibrating Function,
Knocking Function

~~\$189.00~~ \$169.00



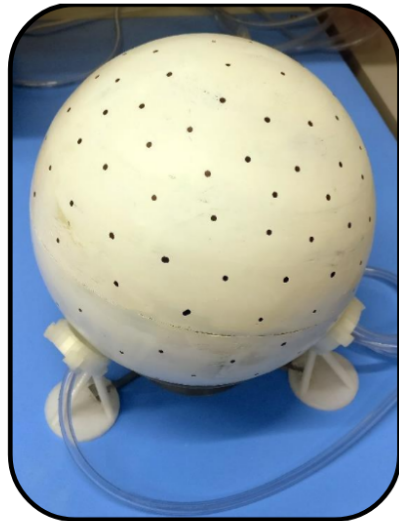
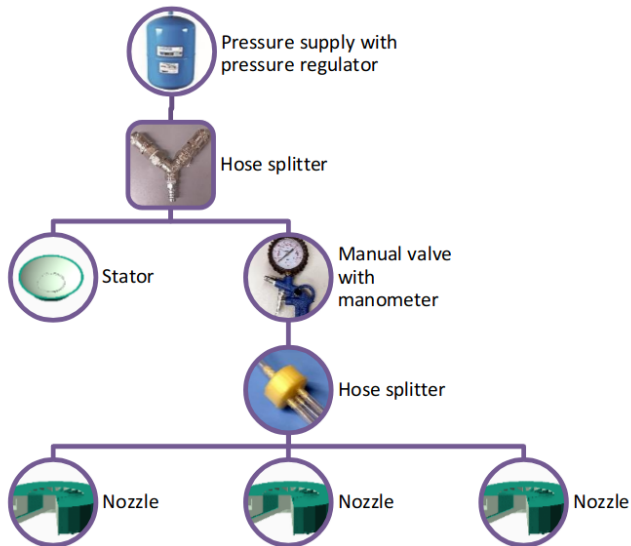
Select options



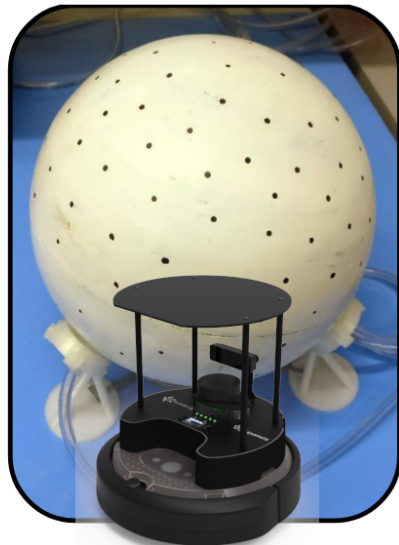
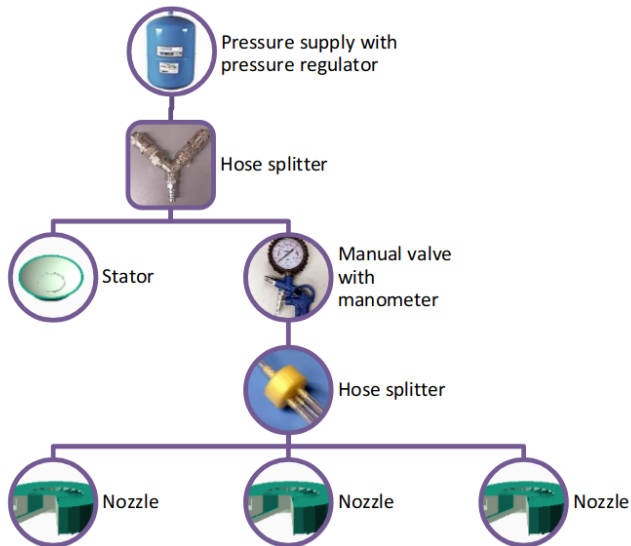
Ceiling Vibrator System V3 Version:
Automatic Noise Sensor Version

~~\$249.00~~ \$189.00

Master's project on spacecraft simulator



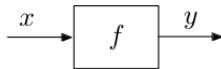
Master's project on spacecraft simulator



Block diagrams

Block diagrams are a notation for looking at systems from an input-output perspective

Block diagram of a function

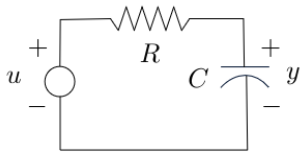


$$f : X \mapsto Y \quad (2)$$

$$y = f(x) \quad (3)$$

Example: RC Circuit

Consider an RC circuit



$$RC\dot{y} + y = u \quad (4)$$

The solution to the ODE is then,

$$y = f(u, y(0)) \quad (5)$$

Thus a valid block diagram is simply,

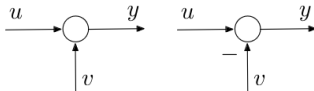


This is a first-order ODE, we need:

- ▶ Initial voltage $y(0)$
- ▶ Input voltage $u(\tau)$ for $\tau \in [0, 1]$

Block diagram algebra

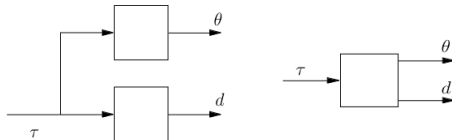
1. Summing junctions: $y = u + v$ or $y = u - v$



2. Multiple inputs: $y = f(u, v)$

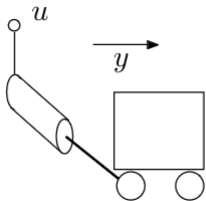


3. Multiple outputs: $\theta = f(\tau)$ and $d = g(\tau)$



Example: 1D Electric Car with a Motor Drive

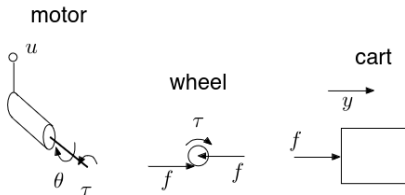
We want the block diagram for position y as a function of motor voltage u



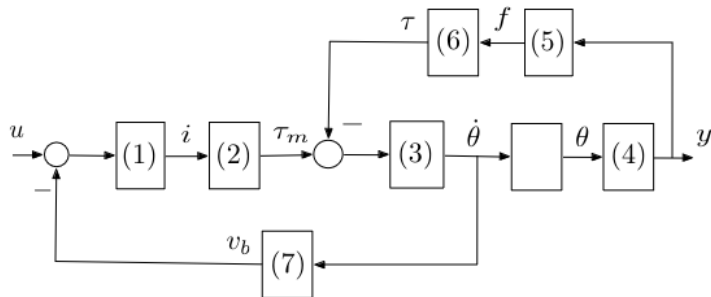
- ▶ Force on car from wheel: $M\ddot{y} = f$
- ▶ Torque on wheel from axle: $\tau = fr$
- ▶ Circuit equation for motor:

$$L\frac{di}{dt} + Ri = u - v_b \quad (6)$$

- ▶ Torque from motor: $\tau_m = Ki$
- ▶ Newton's law for motor shaft: $J\ddot{\theta} = \tau_m - \tau$
- ▶ Back emf from motor rotation: $v_b = K_b\dot{\theta}$
- ▶ Car position from shaft angle: $y = r\theta$



Example: 1D Electric Car with a Motor Drive



$$(1) \quad L \frac{di}{dt} + Ri = u - v_b$$

$$(2) \quad \tau_m = Ki$$

$$(3) \quad J \frac{d\dot{\theta}}{dt} = \tau_m - \tau$$

$$(4) \quad y = r\theta$$

$$(5) \quad f = M\ddot{y}$$

$$(6) \quad \tau = rf$$

$$(7) \quad v_b = K_b \dot{\theta}$$

Transfer function motivation

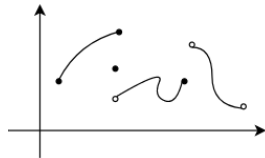
transfer functions are equivalent to linear systems in frequency domain, poles of TFs are related to system behaviour – need laplace transform for this

Laplace Transform

The Laplace transform is given by,

$$\mathcal{F}\{f(t)\} := F(s) = \int_0^{\infty} f(t)e^{-st} dt \quad (7)$$

- ▶ $f(t)$ is **piecewise continuous**



$f(t)$ is continuous except at a countable number of times
 $0 = t_0 < t_1 < t_2 \dots$

- ▶ $f(t)$ is **exponentially bounded**:

$$|f(t)| \leq Me^{at} \quad (8)$$

for some $M \geq 0, a \in \mathbb{R}, t > 0$.

Example: $f(t) = \sin(\omega t)$

Example: $f(t) = \sin(\omega t)$

Example: Derivatives

Properties of the Laplace Transform

► **Linearity:**

$$\alpha f(t) + \beta g(t) \mapsto \alpha F(s) + \beta G(s) \quad (9)$$

► **Convolution:** for causal signals $g(t), h(t) = 0$ for $t < 0$:

$$f(t) := g(t) * h(t) := \int_{-\infty}^{\infty} g(t - \tau)h(\tau)d\tau \quad (10)$$

► **Fourier Transform of Convolution:**

$$F(s) = G(s)H(s) \quad (11)$$

Proof of Convolution Formula

Proof of Convolution Formula

Proof of Convolution Formula

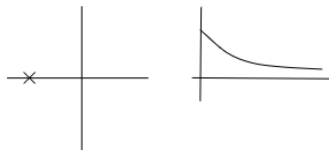
Poles and their influence on system behaviour

Poles and their Laplace Transforms

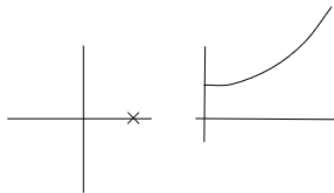
Linear systems are characterized by exponential decays:

$$\dot{x} = -ax \implies x(t) = x(0)e^{-at} \quad (12)$$

$$f(t) = e^{-t}, \quad F(s) = \frac{1}{s+1}$$



$$f(t) = e^t, \quad F(s) = \frac{1}{s-1}$$



Simple poles and what they do under the Laplace transform

- ▶ A single real negative pole corresponds to a decaying exponential. The farther left the pole is, the faster $f(t)$ decays.
- ▶ A single pole at $s = 0$ corresponds to a constant in the time domain.
- ▶ A single real positive pole corresponds to an exponential that blows up. The farther right the pole is, the faster $f(t)$ blows up.
- ▶ A complex conjugate pair of poles with $\Re(s) < 0$ corresponds to a sinusoid with decaying amplitude.
- ▶ A complex conjugate pair of poles on the imaginary axis corresponds to a sinusoid with constant amplitude.
- ▶ A complex conjugate pair of poles with $\Re(s) > 0$ corresponds to a sinusoid with amplitude that blows up.

Poles of multiplicity more than one

- ▶ A double real negative pole corresponds to an amplitude-modified decaying exponential:

$$F(s) = \frac{1}{(s+1)^2} \implies f(t) = te^{-t} \quad (13)$$

- ▶ A double real positive pole corresponds to an amplitude-modified blowing-up exponential:

$$F(s) = \frac{1}{(s-1)^2} \implies f(t) = te^t \quad (14)$$

- ▶ A complex conjugate pair of poles on the imaginary axis of multiplicity two corresponds to a sinusoid with ramp-like amplitude.

$$F(s) = \frac{1}{(s^2+1)^2} \implies f(t) = \frac{1}{2} (\sin t - t \cos t) \quad (15)$$

What about more complicated poles?

We consider **rational functions** (a ratio of polynomials):

$$F(s) = \frac{N(s)}{D(s)} = \frac{(z_1 - s) \cdots (z_n - s)}{(p_1 - s) \cdots (p_m - s)}. \quad (16)$$

Terminology:

- ▶ The roots of the numerator $\{z_i\}$ are called **zeros** of $F(s)$
- ▶ The roots of the denominator $\{p_i\}$ are called **poles** of $F(s)$
- ▶ The number of zeros (or poles) is called the **degree** of $N(s)$ (or $D(s)$)
- ▶ Then, $F(s)$ is
 - ▶ **strictly proper** if $\deg N(s) < \deg D(s)$
 - ▶ **proper** if $\deg N(s) \leq \deg D(s)$
 - ▶ **improper** if $\deg N(s) > \deg D(s)$

What about more complicated poles?

Recall: $f(t)$ is **bounded**: $|f(t)| \leq M$ for some $M \geq 0, t > 0$.

- ▶ If $F(s)$ has no poles in $\Re(s) \geq 0$, then $f(t)$ is bounded
- ▶ If $F(s)$ has no poles in $\Re(s) \geq 0$ except a simple pole at $s = 0$ and/or some simple complex-conjugate pairs of poles on the imaginary axis, then $F(s)$ is bounded.
- ▶ In all other cases, $f(t)$ is unbounded.

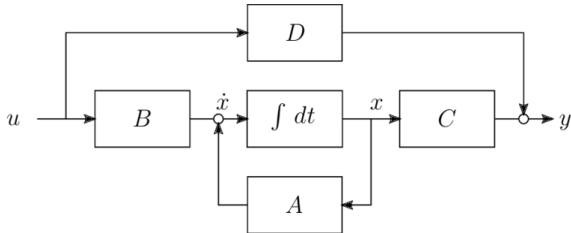
So, why do we care about rational functions?

Motivation for Transfer Functions

Linear systems (time domain) are equivalent to rational functions (frequency domain)

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

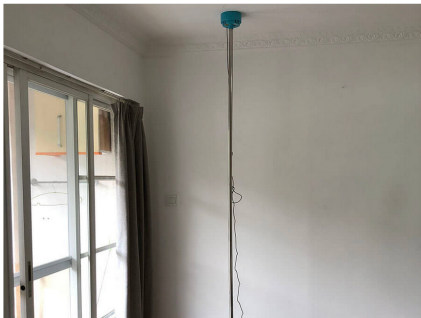


Neighbour Device

Ceiling Vibrator (Thumper): Loud Noisy Upstairs Neighbors Ultimate Solution

You know what?

When I started writing this article, I had just come through a dark time that caused me to suffer from mild depression and neurological debilitation until I got a **ceiling vibrator**. Also, others may call it ceiling thumper, ceiling vibration machine, or ceiling shaker.



Model the device as a mass-spring-damper:

$$\ddot{x} + 2\xi\omega_n\dot{x} + \omega_n^2x = u(t) \quad (17)$$

Goal: Design $u(t)$ to alert neighbour

Measure: Position of spring $y(t) = x(t)$

Want: $G(s)$ such that $Y(s) = G(s)U(s)$

Neighbour Device: State-Space Model

Neighbour Device: Laplace Transform

Transfer Function

Stability

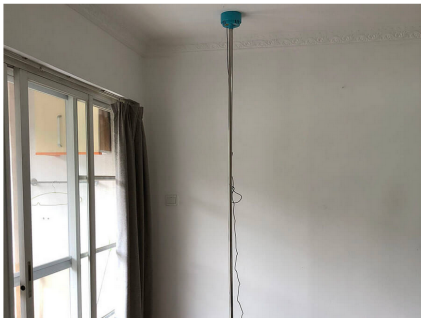
A stable system returns to equilibrium after a perturbation

Neighbour Device

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Model the device as a mass-spring-damper:

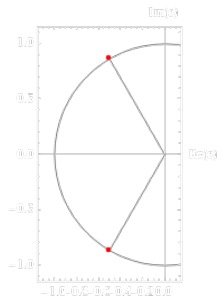
$$\ddot{x} + 2\xi\omega_n\dot{x} + \omega_n^2x = u(t) \quad (18)$$

$$G(s) = \frac{1}{(s - z_1)(s - z_2)} \quad (19)$$

$$z_1 = \frac{1}{2} - i\frac{\sqrt{3}}{2} \quad (20)$$

$$z_2 = \frac{1}{2} + i\frac{\sqrt{3}}{2} \quad (21)$$

Neighbour Device



Model the device as a mass-spring-damper:

$$\ddot{x} + \omega_n \dot{x} + x = u(t) \quad (22)$$

$$G(s) = \frac{1}{(s - z_1)(s - z_2)} \quad (23)$$

$$z_1 = \frac{1}{2} - i\frac{\sqrt{3}}{2} \quad (24)$$

$$z_2 = \frac{1}{2} + i\frac{\sqrt{3}}{2} \quad (25)$$

Stability

The linear system

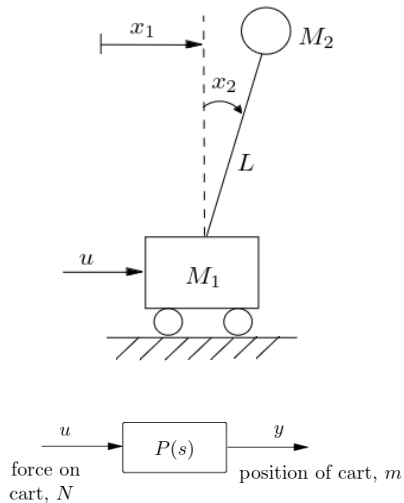
$$\dot{x} = Ax + Bu \quad (26)$$

$$y = Cx \quad (27)$$

is:

- ▶ **Asymptotically stable** if with $u = 0$ and arbitrary $x(t)$, $x(t) \rightarrow 0$
- ▶ **BIBO stable** if with $x(0) = 0$, $y(t)$ is bounded for every bounded $u(t)$
- ▶ The linear system is AS if and only if all the eigenvalues of A have negative real part

Feedback Loop: A Motivation



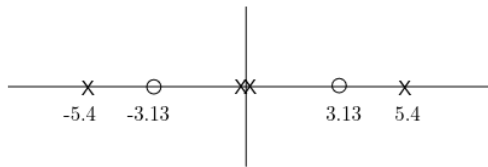
$$\dot{x} = Ax + Bu$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -19.6 & 0 & 0 \\ 0 & 29.4 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$$

Feedback Loop: A Motivation



$$P(s) = \frac{s^2 - 9.8}{s^2(s^2 - 29.4)} \quad (28)$$

Observations:

- ▶ RHP poles: **unstable**
- ▶ RHP zeros: **non-minimum phase**

Feedback Loop: A Motivation



Naive Solution: Design controller $K(s)$ so that it cancels the unstable poles

$$K(s) = s^2 - 29.4 \implies P(s)K(s) = (s^2 - 29.4) \frac{s^2 - 9.8}{s^2(s^2 - 29.4)} = \frac{s^2 - 9.8}{s^2} \quad (29)$$

Boom, the system is stable

Thanks for the course!



Good luck with your master's thesis!

Okay, but actually...

In theory this works, in practice it's impossible

What actually happens:

- ▶ You design $K(s)$
- ▶ $K(s)$ is a physical and/or digital system
- ▶ You have precision constraints, so

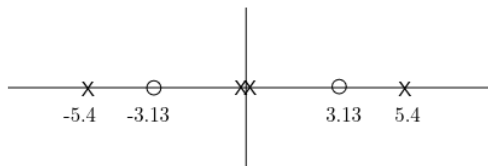
$$K(s) = s^2 - 29.4 + \epsilon \quad (30)$$

- ▶ So,

$$P(s)K(s) = \frac{(s^2 - 29.4 + \epsilon)(s^2 - 9.8)}{s^2(s^2 - 29.4)} \quad (31)$$

- ▶ Fantastic job, you've now added additional non-minimum phase behaviour on top of the instability
- ▶ You get fired (ok, maybe not in Norway)

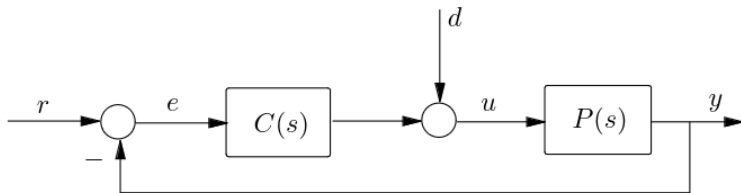
So what do you do?



$$P(s) = \frac{s^2 - 9.8}{s^2(s^2 - 29.4)} \quad (32)$$

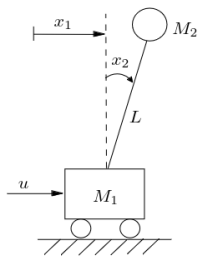
- ▶ You shift the poles into the left-hand-plane
- ▶ This can be accomplished with feedback

Closed-Loop Control



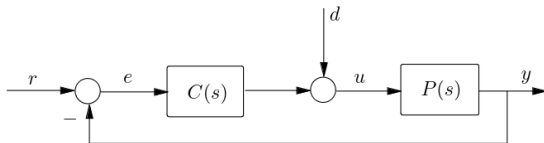
- ▶ Supply a **reference** r (eg., $\theta = 90^\circ$)
- ▶ Define the **error** as $e := r - y$
- ▶ Try to get $e \rightarrow 0$ using a **controller** $C(s)$

Feedback Loop: A Motivation

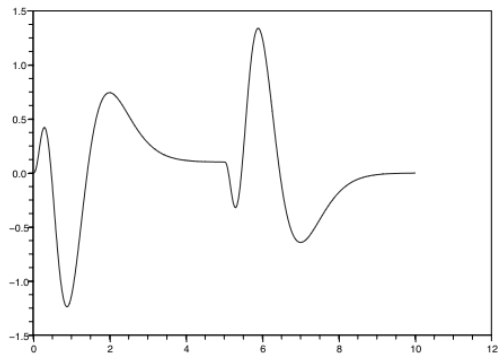
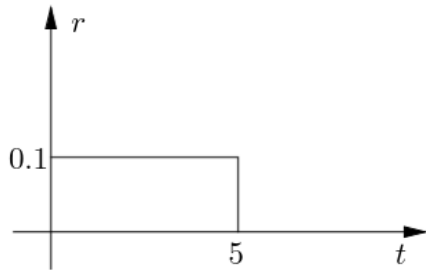


$$P(s) = \frac{s^2 - 9.8}{s^2(s^2 - 29.4)} \quad (33)$$

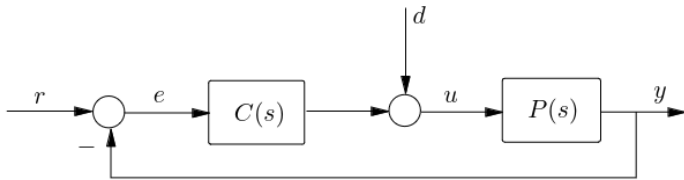
$$C(s) = \frac{10395s^3 + 54126s^2 - 13375s - 6687}{s^4 + 32s^3 + 477s^2 - 5870s - 22170} \quad (34)$$



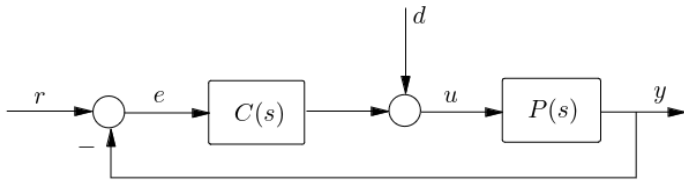
Matlab Example



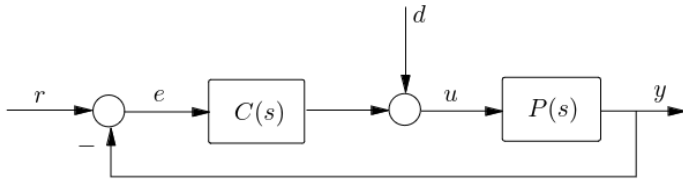
Standard Feedback Control



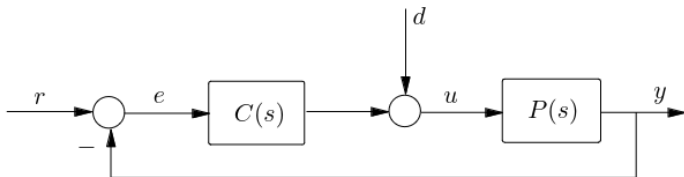
Standard Feedback Control



Standard Feedback Control



Theorem of Stability for Feedback Control



Theorem

The closed-loop feedback system is stable if and only if the zeros of the **characteristic polynomial**

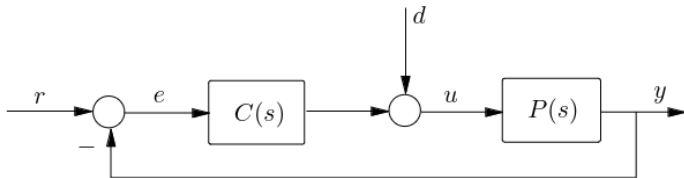
$$D_p D_c + N_p N_c \quad (35)$$

are all in the open left-hand plane.

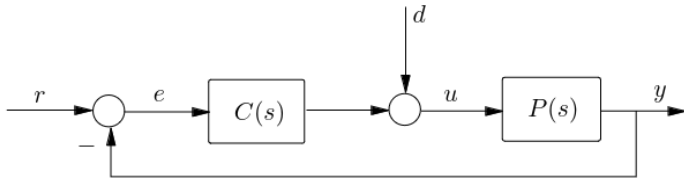
Example 1

Example 2

We saw the transfer function from e to u ...



All the Transfer Functions



$$\begin{bmatrix} E \\ U \end{bmatrix} = \begin{bmatrix} \frac{1}{1+PC} & -\frac{P}{1+PC} \\ \frac{C}{1+PC} & \frac{1}{1+PC} \end{bmatrix} \begin{bmatrix} R \\ D \end{bmatrix} \quad (36)$$

Eg., transfer function from D to E is:

$$-\frac{P}{1+PC} \quad (37)$$

Reference Tracking: Motivation

Reference Tracking: Final Value

Theorem

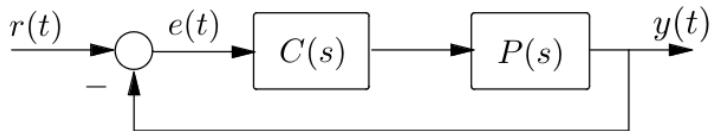
Consider a signal $f(t)$ with (rational, strictly proper) Laplace transform $F(s)$

- ▶ If $F(s)$ has no poles in $\Re(s) \geq 0$, then $f(t)$ has a final value: $\lim_{t \rightarrow \infty} f(t) = 0$
- ▶ If $F(s)$ has no poles in $\Re(s) \geq 0$ except a simple pole at $s = 0$, then

$$\lim_{s \rightarrow 0} sF(s) \tag{38}$$

- ▶ In all other cases, f is unstable, so $\lim_{t \rightarrow \infty} f(t) = \infty$

Reference Tracking



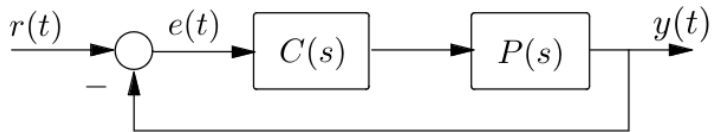
$$P(s) = \frac{1}{s + 1}$$

$$C(s) = 1$$

$$R(s) = \frac{100}{s}$$

This is a simple model for cruise control

Reference Tracking



Next Time:

- ▶ More on state space systems & transfer function equivalence
- ▶ Stability tools: Bode plots, Nyquist diagrams, gain/phase margins
- ▶ Loop shaping